

MT209: Fundamentals of Metal Extraction

Class Test - 2

Total Marks [15]

Date: ~~17~~¹⁰th Apr 2026Time: ~~20:00 to 20:50~~~~09:00 to 09:50 am~~

1. Differentiate the following: [03]

- Hot Metal as output product of iron making (blast furnace) and Hot Metal as input raw material of steel making (BOF)
- Iron making slag obtained from Blast Furnace and Steel making slag obtained from BOF
- Oxygen required for iron making in blast furnace and oxygen required for BOF

2. For a hypothetical BOF steelmaking process consider the following data: [05]

Hot Metal : 3.6% C, 1.4% Si, 0.55% Mn and rest Fe

Scrap quantity added: 10% of Hot Metal

Scrap composition: same as steel composition

Steel composition at tapping: 0.12% C, and rest Fe

Slag consist of 18% of FeO with basicity (wt. CaO/ wt. SiO₂) = 3.2Assume gas coming out consist only CO₂Assume Flux composition: α % CaO, and rest MgO

32%

8%

 α is last two digits of student ID

i. Calculate only using mass balance:

- Mass of Hot Metal needed (in kg/TLS)
- Mass of Slag generated (in kg/TLS)
- Volume of CO₂ generated (in Nm³/TLS)
- Volume of total O₂ needed (in Nm³/TLS)
- Mass of Flux needed/TLS

- List the possible oxidation reactions considered for above calculations
- Calculate only using mass balance:

[02]
[05]

Amount of Elements entering or leaving BOF	(in kg-mol)
Fe	
C	
Si	
Mn	
O	

$$\begin{array}{r} 3.6 \\ + 1.4 \\ \hline 5.0 \end{array} \quad \begin{array}{r} 100.00 \\ - 5.55 \\ \hline 94.45 \end{array} \quad \begin{array}{r} 100.00 \\ - 0.12 \\ \hline 99.88 \end{array}$$

$$1000 \times \frac{1.4}{100}$$

$$\frac{14 \times 3.2}{28}$$

$$\frac{44.8}{1}$$

MgO removed =

$$\begin{array}{r} 18 \\ + 16 \\ \hline 34 \\ + 54 \\ \hline 88 \end{array}$$